Type MS · Turning sums of trig terms into products, and products into sums

Answer 1 of 8 · Question 10 part b · 3 marks

2020 NOV · Paper 02 · extract 1 of 3 · find part b in this mark scheme extract

10(a)	$\cos(A+B)+\cos(A-B)=\cos A\cos B-\sin A\sin B+\cos A\cos B+\sin A\sin B$	M1
	$=2\cos A\cos B$ *	A1 cso (2)
(b)	Let $A+B=P,\quad A-B=Q \Rightarrow A=\frac{1}{2}(P+Q),\quad B=\frac{1}{2}(P-Q)$	M1
	As $\cos(A+B)+\cos(A-B)=2\cos A\cos B$	
	$\cos P+\cos Q=2\cos\frac{1}{2}(P+Q)\cos\frac{1}{2}(P-Q)$ *	M1A1cso (3)
ALT		
	Let $A=\frac{1}{2}(P+Q),\quad B=\frac{1}{2}(P-Q)\Rightarrow A+B=P,\ A-B=Q$	M1
	As $2\cos A\cos B=\cos(A+B)+\cos(A-B)$	
	$2\cos\frac{1}{2}(P+Q)\cos\frac{1}{2}(P-Q)=\cos P+\cos Q$ *	M1A1cso
(c)	$\cos 5\theta+\cos 7\theta=2\cos 6\theta\cos\theta=0$	M1
	$\cos 6\theta=0\Rightarrow 6\theta=\frac{\pi}{2},\frac{3\pi}{2},\frac{5\pi}{2}\Rightarrow\theta=\frac{\pi}{12},\frac{\pi}{4}\ \left(\text{or}\ \frac{3\pi}{12}\right),\frac{5\pi}{12}$	A1A1
	$\cos\theta=0\Rightarrow\theta=\frac{\pi}{2}$	A1 (4)
(d)	$\cos 8x+2\cos 6x+\cos 4x=(\cos 8x+\cos 6x)+(\cos 6x+\cos 4x)$	
	$=2\cos 7x\cos x+2\cos 5x\cos x$	M1
	$=2\cos x(\cos 7x+\cos 5x)=2\cos x\times 2(\cos 6x\cos x),\ =4\cos 6x\cos^2 x$ *	dM1,A1cso (3)
ALT 1	$\cos 8x+2\cos 6x+\cos 4x=(\cos 8x+\cos 4x)+2\cos 6x$	
	$=2\cos 6x\cos 2x+2\cos 6x$	M1
	$=2\cos 6x(\cos 2x+1)=2\cos 6x(2\cos^2 x-1+1)=4\cos 6x\cos^2 x$ *	dM1A1cso (3)
ALT 2	Working from right to left	
	$4\cos 6x\cos^2 x=4\cos 6x\times\frac{1}{2}(\cos 2x+1)=2\cos 6x\cos 2x+2\cos 6x$	M1
	$=\cos 8x+\cos 4x+2\cos 6x$ *	dM1A1cso (3)

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(e)	$\int_0^{\frac{\pi}{3}} \cos 6x \cos^2 x \, dx = \frac{1}{4}\int_0^{\frac{\pi}{3}} (\cos 8x + 2\cos 6x + \cos 4x) \, dx$	M1	
	$= \frac{1}{4}\left[\frac{1}{8}\sin 8x + \frac{1}{3}\sin 6x + \frac{1}{4}\sin 4x\right]_0^{\frac{\pi}{3}}$	A1	
	$= \frac{1}{4}\left[\frac{1}{8}\sin\frac{8\pi}{3} + \frac{1}{3}\sin 2\pi + \frac{1}{4}\sin\frac{4\pi}{3} - 0\right] = -\frac{\sqrt{3}}{64}$ (-0.02706... scores M1A0)	dM1A1	(4)
			[16]
ALT	For the first M mark Award M1 only when the integrand has been changed (by a valid method) to a function which can be integrated $\int_0^{\frac{\pi}{3}} \cos 6x \cos^2 x \, dx = \int_0^{\frac{\pi}{3}} \cos 6x \times \frac{1}{2}(\cos 2x + 1) \, dx$		
	$= \frac{1}{2}\int_0^{\frac{\pi}{3}} (\cos 6x \times \cos 2x + \cos 6x) \, dx = \frac{1}{2}\int_0^{\frac{\pi}{3}} \frac{1}{2}(\cos 8x + \cos 4x + 2\cos 6x) \, dx$	M1	
	Rest as above.		

(a)	
M1	Use the standard formulae. This is a "show that" question so these formulae must be written out in full. Both sides of the result must be seen although the working may appear between them as seen in the scheme.
A1cso	Final **given** result. Do not award if the expansions shown are not in the correct order (this suggests incorrect signs in the formulae).
(b)	
M1	Use $A+B=P,\quad A-B=Q$ to obtain A and B in terms of P and Q
M1	Substitute in the result from (a) to obtain an identity in P and Q only
A1cso	Correct result reached with no errors seen
ALT	Working right to left: Notes similar to above
(c)	
M1	Use the result from (b) to show that $(2)\cos 6\theta \cos(\pm\theta) = 0$
A1	Obtain one correct solution of $\cos 6\theta = 0$. Allow if in decimal form but must be radians.
A1	Two further correct solutions **and no more within the range.** If any of the 3 solutions of $\cos 6\theta = 0$ is not exact do not award this mark.
A1	State the solution of $\cos(\pm\theta) = 0$. Must be exact unless this is penalised above.
NB	If answers are given in degrees deduct 2A marks from any that would otherwise have been given. (Answers in degrees but then changed to radians are acceptable – mark the radian answers.) Ignore extra answers outside the stated range – any within are incorrect.
(d)	
M1	Use the result from (a) or (b) once. Allow if one or both "2"s are missing
dM1	Use the result from (a) or (b) again. Must include both "2"s this time. Depends on previous M mark.
A1cso	Reach the **given** result with no errors seen

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ALT 1	
M1	Use the result from (a) or (b). Allow if the "2" is missing.
dM1	Factorise and use the *correct* double angle formula on cos2x Depends on the previous M mark
A1cso	Reach the **given** result with no errors seen
ALT 2	
M1	Use the *correct* half angle formula on $\cos^2 x$
dM1A1	M1: Use the result from (a) or (b)
cso	A1: reach the **given** result with no errors seen
(e)M1	Obtain a function which can be integrated **either** by using the **given** result from (d) **OR**. deriving the same result. Allow if 1/4 is missing. (Integration by Parts – send to review)
A1	Correct integration (must have included the ¼) Limits not needed for these 2 marks.
dM1	Correct use of the given limits. All sines are 0 at the lower limit so these need not be shown
A1	Correct final answer which must be **exact and stated as a single fraction**.

Type MS · Turning sums of trig terms into products, and products into sums

Answer 2 of 8 · Question 10 part a · 2 marks

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10(a)	$\cos(A+B)+\cos(A-B)=\cos A\cos B-\sin A\sin B+\cos A\cos B+\sin A\sin B$	M1
	$=2\cos A\cos B$ *	A1 cso (2)
(b)	Let $A+B=P,\quad A-B=Q \Rightarrow A=\frac{1}{2}(P+Q),\quad B=\frac{1}{2}(P-Q)$	M1
	As $\cos(A+B)+\cos(A-B)=2\cos A\cos B$	
	$\cos P+\cos Q=2\cos\frac{1}{2}(P+Q)\cos\frac{1}{2}(P-Q)$ *	M1A1cso (3)
ALT		
	Let $A=\frac{1}{2}(P+Q),\quad B=\frac{1}{2}(P-Q)\Rightarrow A+B=P,\ A-B=Q$	M1
	As $2\cos A\cos B=\cos(A+B)+\cos(A-B)$	
	$2\cos\frac{1}{2}(P+Q)\cos\frac{1}{2}(P-Q)=\cos P+\cos Q$ *	M1A1cso
(c)	$\cos 5\theta+\cos 7\theta=2\cos 6\theta\cos\theta=0$	M1
	$\cos 6\theta=0\Rightarrow 6\theta=\frac{\pi}{2},\frac{3\pi}{2},\frac{5\pi}{2}\Rightarrow\theta=\frac{\pi}{12},\frac{\pi}{4}\ \left(\text{or}\ \frac{3\pi}{12}\right),\frac{5\pi}{12}$	A1A1
	$\cos\theta=0\Rightarrow\theta=\frac{\pi}{2}$	A1 (4)
(d)	$\cos 8x+2\cos 6x+\cos 4x=(\cos 8x+\cos 6x)+(\cos 6x+\cos 4x)$	
	$=2\cos 7x\cos x+2\cos 5x\cos x$	M1
	$=2\cos x(\cos 7x+\cos 5x)=2\cos x\times 2(\cos 6x\cos x),\ =4\cos 6x\cos^2 x$ *	dM1,A1cso (3)
ALT 1	$\cos 8x+2\cos 6x+\cos 4x=(\cos 8x+\cos 4x)+2\cos 6x$	
	$=2\cos 6x\cos 2x+2\cos 6x$	M1
	$=2\cos 6x(\cos 2x+1)=2\cos 6x(2\cos^2 x-1+1)=4\cos 6x\cos^2 x$ *	dM1A1cso (3)
ALT 2	Working from right to left	
	$4\cos 6x\cos^2 x=4\cos 6x\times\frac{1}{2}(\cos 2x+1)=2\cos 6x\cos 2x+2\cos 6x$	M1
	$=\cos 8x+\cos 4x+2\cos 6x$ *	dM1A1cso (3)

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Answer 2 of 8 · Question 10 part a · 2 marks

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(e)	$\int_0^{\frac{\pi}{3}} \cos 6x \cos^2 x \, dx = \frac{1}{4}\int_0^{\frac{\pi}{3}} (\cos 8x + 2\cos 6x + \cos 4x) \, dx$	M1	
	$= \frac{1}{4}\left[\frac{1}{8}\sin 8x + \frac{1}{3}\sin 6x + \frac{1}{4}\sin 4x\right]_0^{\frac{\pi}{3}}$	A1	
	$= \frac{1}{4}\left[\frac{1}{8}\sin\frac{8\pi}{3} + \frac{1}{3}\sin 2\pi + \frac{1}{4}\sin\frac{4\pi}{3} - 0\right] = -\frac{\sqrt{3}}{64}$ (-0.02706... scores M1A0)	dM1A1	(4)
			[16]
ALT	For the first M mark Award M1 only when the integrand has been changed (by a valid method) to a function which can be integrated $\int_0^{\frac{\pi}{3}} \cos 6x \cos^2 x \, dx = \int_0^{\frac{\pi}{3}} \cos 6x \times \frac{1}{2}(\cos 2x + 1) \, dx$		
	$= \frac{1}{2}\int_0^{\frac{\pi}{3}} (\cos 6x \times \cos 2x + \cos 6x) \, dx = \frac{1}{2}\int_0^{\frac{\pi}{3}} \frac{1}{2}(\cos 8x + \cos 4x + 2\cos 6x) \, dx$	M1	
	Rest as above.		

(a)	
M1	Use the standard formulae. This is a "show that" question so these formulae must be written out in full. Both sides of the result must be seen although the working may appear between them as seen in the scheme.
A1cso	Final **given** result. Do not award if the expansions shown are not in the correct order (this suggests incorrect signs in the formulae).
(b)	
M1	Use $A+B=P,\quad A-B=Q$ to obtain A and B in terms of P and Q
M1	Substitute in the result from (a) to obtain an identity in P and Q only
A1cso	Correct result reached with no errors seen
ALT	Working right to left: Notes similar to above
(c)	
M1	Use the result from (b) to show that $(2)\cos 6\theta \cos(\pm\theta) = 0$
A1	Obtain one correct solution of $\cos 6\theta = 0$. Allow if in decimal form but must be radians.
A1	Two further correct solutions **and no more within the range.** If any of the 3 solutions of $\cos 6\theta = 0$ is not exact do not award this mark.
A1	State the solution of $\cos(\pm\theta) = 0$. Must be exact unless this is penalised above.
NB	If answers are given in degrees deduct 2A marks from any that would otherwise have been given. (Answers in degrees but then changed to radians are acceptable – mark the radian answers.) Ignore extra answers outside the stated range – any within are incorrect.
(d)	
M1	Use the result from (a) or (b) once. Allow if one or both "2"s are missing
dM1	Use the result from (a) or (b) again. Must include both "2"s this time. Depends on previous M mark.
A1cso	Reach the **given** result with no errors seen

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Answer 2 of 8 · Question 10 part a · 2 marks

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ALT 1	
M1	Use the result from (a) or (b). Allow if the "2" is missing.
dM1	Factorise and use the *correct* double angle formula on cos2x Depends on the previous M mark
A1cso	Reach the **given** result with no errors seen
ALT 2	
M1	Use the *correct* half angle formula on $\cos^2 x$
dM1A1	M1: Use the result from (a) or (b)
cso	A1: reach the **given** result with no errors seen
(e)M1	Obtain a function which can be integrated **either** by using the **given** result from (d) **OR**. deriving the same result. Allow if 1/4 is missing. (Integration by Parts – send to review)
A1	Correct integration (must have included the ¼) Limits not needed for these 2 marks.
dM1	Correct use of the given limits. All sines are 0 at the lower limit so these need not be shown
A1	Correct final answer which must be **exact and stated as a single fraction**.

Type MS · Turning sums of trig terms into products, and products into sums

Answer 3 of 8 · Question 10 part d · 3 marks

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10(a)	$\cos(A+B)+\cos(A-B)=\cos A\cos B-\sin A\sin B+\cos A\cos B+\sin A\sin B$	M1
	$=2\cos A\cos B$ *	A1 cso (2)
(b)	Let $A+B=P,\quad A-B=Q \Rightarrow A=\frac{1}{2}(P+Q),\quad B=\frac{1}{2}(P-Q)$	M1
	As $\cos(A+B)+\cos(A-B)=2\cos A\cos B$	
	$\cos P+\cos Q=2\cos\frac{1}{2}(P+Q)\cos\frac{1}{2}(P-Q)$ *	M1A1cso (3)
ALT		
	Let $A=\frac{1}{2}(P+Q),\quad B=\frac{1}{2}(P-Q)\Rightarrow A+B=P,\ A-B=Q$	M1
	As $2\cos A\cos B=\cos(A+B)+\cos(A-B)$	
	$2\cos\frac{1}{2}(P+Q)\cos\frac{1}{2}(P-Q)=\cos P+\cos Q$ *	M1A1cso
(c)	$\cos 5\theta+\cos 7\theta=2\cos 6\theta\cos\theta=0$	M1
	$\cos 6\theta=0\Rightarrow 6\theta=\frac{\pi}{2},\frac{3\pi}{2},\frac{5\pi}{2}\Rightarrow\theta=\frac{\pi}{12},\frac{\pi}{4}\ \left(\text{or}\ \frac{3\pi}{12}\right),\frac{5\pi}{12}$	A1A1
	$\cos\theta=0\Rightarrow\theta=\frac{\pi}{2}$	A1 (4)
(d)	$\cos 8x+2\cos 6x+\cos 4x=(\cos 8x+\cos 6x)+(\cos 6x+\cos 4x)$	
	$=2\cos 7x\cos x+2\cos 5x\cos x$	M1
	$=2\cos x(\cos 7x+\cos 5x)=2\cos x\times 2(\cos 6x\cos x),=4\cos 6x\cos^2 x$ *	dM1,A1cso (3)
ALT 1	$\cos 8x+2\cos 6x+\cos 4x=(\cos 8x+\cos 4x)+2\cos 6x$	
	$=2\cos 6x\cos 2x+2\cos 6x$	M1
	$=2\cos 6x(\cos 2x+1)=2\cos 6x(2\cos^2 x-1+1)=4\cos 6x\cos^2 x$ *	dM1A1cso (3)
ALT 2	Working from right to left	
	$4\cos 6x\cos^2 x=4\cos 6x\times\frac{1}{2}(\cos 2x+1)=2\cos 6x\cos 2x+2\cos 6x$	M1
	$=\cos 8x+\cos 4x+2\cos 6x$ *	dM1A1cso (3)

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Answer 3 of 8 · Question 10 part d · 3 marks

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(e)	$\int_0^{\frac{\pi}{3}} \cos 6x \cos^2 x \, dx = \frac{1}{4}\int_0^{\frac{\pi}{3}} (\cos 8x + 2\cos 6x + \cos 4x) \, dx$	M1	
	$= \frac{1}{4}\left[\frac{1}{8}\sin 8x + \frac{1}{3}\sin 6x + \frac{1}{4}\sin 4x\right]_0^{\frac{\pi}{3}}$	A1	
	$= \frac{1}{4}\left[\frac{1}{8}\sin\frac{8\pi}{3} + \frac{1}{3}\sin 2\pi + \frac{1}{4}\sin\frac{4\pi}{3} - 0\right] = -\frac{\sqrt{3}}{64}$ (-0.02706... scores M1A0)	dM1A1	(4)
			[16]
ALT	For the first M mark Award M1 only when the integrand has been changed (by a valid method) to a function which can be integrated $\int_0^{\frac{\pi}{3}} \cos 6x \cos^2 x \, dx = \int_0^{\frac{\pi}{3}} \cos 6x \times \frac{1}{2}(\cos 2x + 1) \, dx$		
	$= \frac{1}{2}\int_0^{\frac{\pi}{3}} (\cos 6x \times \cos 2x + \cos 6x) \, dx = \frac{1}{2}\int_0^{\frac{\pi}{3}} \frac{1}{2}(\cos 8x + \cos 4x + 2\cos 6x) \, dx$	M1	
	Rest as above.		

(a)	
M1	Use the standard formulae. This is a "show that" question so these formulae must be written out in full. Both sides of the result must be seen although the working may appear between them as seen in the scheme.
A1cso	Final **given** result. Do not award if the expansions shown are not in the correct order (this suggests incorrect signs in the formulae).
(b)	
M1	Use $A+B=P,\quad A-B=Q$ to obtain A and B in terms of P and Q
M1	Substitute in the result from (a) to obtain an identity in P and Q only
A1cso	Correct result reached with no errors seen
ALT	Working right to left: Notes similar to above
(c)	
M1	Use the result from (b) to show that $(2)\cos 6\theta \cos(\pm\theta) = 0$
A1	Obtain one correct solution of $\cos 6\theta = 0$. Allow if in decimal form but must be radians.
A1	Two further correct solutions **and no more within the range.** If any of the 3 solutions of $\cos 6\theta = 0$ is not exact do not award this mark.
A1	State the solution of $\cos(\pm\theta) = 0$. Must be exact unless this is penalised above.
NB	If answers are given in degrees deduct 2A marks from any that would otherwise have been given. (Answers in degrees but then changed to radians are acceptable – mark the radian answers.) Ignore extra answers outside the stated range – any within are incorrect.
(d)	
M1	Use the result from (a) or (b) once. Allow if one or both "2"s are missing
dM1	Use the result from (a) or (b) again. Must include both "2"s this time. Depends on previous M mark.
A1cso	Reach the **given** result with no errors seen

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Answer 3 of 8 · Question 10 part d · 3 marks

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ALT 1	
M1	Use the result from (a) or (b). Allow if the "2" is missing.
dM1	Factorise and use the *correct* double angle formula on cos2x Depends on the previous M mark
A1cso	Reach the **given** result with no errors seen
ALT 2	
M1	Use the *correct* half angle formula on $\cos^2 x$
dM1A1	M1: Use the result from (a) or (b)
cso	A1: reach the **given** result with no errors seen
(e)M1	Obtain a function which can be integrated **either** by using the **given** result from (d) **OR**. deriving the same result. Allow if 1/4 is missing. (Integration by Parts – send to review)
A1	Correct integration (must have included the ¼) Limits not needed for these 2 marks.
dM1	Correct use of the given limits. All sines are 0 at the lower limit so these need not be shown
A1	Correct final answer which must be **exact and stated as a single fraction**.

Type MS · Turning sums of trig terms into products, and products into sums

Answer 4 of 8 · Question 10 part a · 3 marks

2022 JUN · Paper 01 (regional R) · extract 1 of 4 · find part a in this mark scheme extract

Number		
10 (a)	$\sin(A+B)+\sin(A-B)=\sin A\cos B+\sin B\cos A+\sin A\cos B-\sin B\cos A$ $=2\sin A\cos B$	M1
	$[\sin(A+B)+\sin(A-B)=\sin 5x+\sin 3x]$ $Ax+Bx=5x$ $Ax-Bx=3x$	M1
	$\Rightarrow A=4x,\quad B=x$	
	$\Rightarrow \sin(4x+x)+\sin(4x-x)=\sin 5x+\sin 3x=2\sin 4x\cos x$	A1 cso [3]
	ALT	
	$\sin(A+B)-\sin(A-B)=\sin A\cos B+\sin B\cos A-(\sin A\cos B-\sin B\cos A)$ $=2\sin B\cos A$	[M1
	$[\sin(A+B)+\sin(A-B)=\sin(A+B)-\sin[-(A-B)]=\sin 5x+\sin 3x]$ $\Rightarrow A=x,\quad B=4x$	M1
	$\Rightarrow \sin(x+4x)-\sin(x-4x)=\sin 5x-\sin(-3x)=\sin 5x+\sin 3x=2\sin 4x\cos x$ $[-\sin(-3x)=+\sin 3x]$	A1cso]
(b)	**Intersections on x-axis** $6\sin x\cos x=0\Rightarrow \sin 4x=0,\quad \cos x=0$	M1
	from $\sin 4x=0\Rightarrow 4x=0,\pi,2\pi,3\pi,4\pi\rightarrow x=0,\frac{\pi}{4},\frac{\pi}{2},\frac{3\pi}{2},2\pi$ **and**	M1
	from $\cos x=0\Rightarrow x=0,\frac{\pi}{2},\frac{3\pi}{2}$ $\Rightarrow x=0,\frac{\pi}{4},\frac{\pi}{2}$	A1
	Integration $(3)\int(\sin 5x+\sin 3x)\,dx=(3)\left(-\frac{\cos 5x}{5}-\frac{\cos 3x}{3}\right)$	M1 A1 ft
	Limits $\int_0^{\frac{\pi}{4}}(\sin 5x+\sin 3x)\,dx\pm\int_{\frac{\pi}{4}}^{\frac{\pi}{2}}(\sin 5x+\sin 3x)\,dx$	M1
	Substitution of values	

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	$3\left\{\left[\left(-\frac{\cos 5\left(\frac{\pi}{4}\right)}{5}-\frac{\cos 3\left(\frac{\pi}{4}\right)}{3}\right)-\left(-\frac{\cos 5(0)}{5}-\frac{\cos 3(0)}{3}\right)\right]-\left[\left(-\frac{\cos 5\left(\frac{\pi}{2}\right)}{5}-\frac{\cos 3\left(\frac{\pi}{2}\right)}{3}\right)-\left(-\frac{\cos 5\left(\frac{\pi}{4}\right)}{5}-\frac{\cos 3\left(\frac{\pi}{4}\right)}{3}\right)\right]\right\}$	ddM1
	$= 2.73137... + \lvert -1.313170... \rvert = 3.8267... \approx 3.86$	A1 cao [8]
		Total 11 marks

Part	Mark	Additional Guidance
(a)	M1	Use of both correct trig identities [either adding or subtracting], allow any two distinct letters or symbols and simplify to reach either $2\sin A\cos B$ or $2\sin B\cos A$ Allow use of $4x$ and x in place of A and B
	M1	Finds or deduces that $A = 4x, \quad B = x$ Or if they use the **ALT** scheme then $A = x, \quad B = 4x$ This may be implied by the next step or just written down.
	A1cso	For a final correct statement with no errors seen.
(b)	M1	(Placing the equation = 0) and arriving at one of $\sin 4x = 0$ ***or*** $\cos x = 0$
	M1	At least any **one** correct value for x from $\sin 4x = 0$ **and** at least any **one** correct value for x from $\cos x = 0$
	A1	Correctly identifies the required intersections of the curve with the x-axis either explicitly here or used in limits in following work, 0 need not be explicitly stated.
	Note: Award for these values seen without working, and allow working here in degrees.	
	M1	**Integration** An attempt at integration of both terms from their part (a) to an expression of the form $\pm\frac{\cos 5x}{5}\pm\frac{\cos 3x}{3}$ Ignore: • the presence or absence of the multiplier of 3 for this mark • any limits • incorrect notation
	A1	Fully correct integration. Again, ignore the presence/absence of the multiplier of 3. $(3)\left(-\frac{\cos 5x}{5}-\frac{\cos 3x}{3}\right)$
	M1	**Limits** Candidate clearly shows the integration of the curve needs to be done in two parts, where their limits are explicitly shown and come from the limits identified from solving the equation $6\sin x\cos x = 0$.

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		As a minimum we need to see two integrals of the function (this can either be written in the original form or the changed form). Their limits must be clearly shown for this mark (following through their solutions) and the upper limit on the first integral must be equal to the lower limit on the second. The candidate does not need to show that the modulus of the second integral must be taken for this mark. Ignore the presence/absence of the multiplier of 3.
	ddM1	Substitution of all limits –dependent on the previous two method marks, allow slips in simplification. The substitution must be correct. If you see any of the decimal answers shown in the working column or any of $\frac{8+4\sqrt{2}}{5}$ (1st integral) or $\frac{4\sqrt{2}}{5}$ (2nd integral) or $\frac{8+8\sqrt{2}}{5}$ (sum of integrals) award this mark. **NOTE: They must consider the modulus of the second integral, and the multiplier of 3 must also be taken into account for this mark.**
	A1cao	For awrt 3.86

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Answer 4 of 8 · Question 10 part a · 3 marks

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Question number	Scheme	Marks

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Answer 5 of 8 · Question 6 part a · 2 marks

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number		
6 (a)	$(\sin A\cos B+\cos A\sin B)+(\sin A\cos B-\cos A\sin B)$ $=2\sin A\cos B$ *	M1 A1 cso (2)
(b)	$\sin 8x+\sin 6x$	B1 (1)
(c)	$3\int_0^{\frac{\pi}{4}}(\sin 8x+\sin 6x)\,dx$ $=(3)\left[-\frac{1}{8}\cos 8x-\frac{1}{6}\cos 6x\right]_0^{\frac{\pi}{4}}$ $=(3)\left[\left(-\frac{1}{8}-0\right)-\left(-\frac{1}{8}-\frac{1}{6}\right)\right]$ $=\frac{1}{2}$ cao oe	M1 A1 M1 A1cao (4) [7]

Part	Mark	Additional Guidance
(a)	M1	Correct expression show.
	A1	cso
(b)	B1	For the expression shown
(c)	M1	$k\int_0^{\frac{\pi}{4}}(\sin 8x+\sin 6x)\,dx$ $k\neq 0$ or 1 k must be an integer. This mark can be awarded if the limits aren't seen on the integral.
	A1	Correctly integrated, the 3 and the limits need not be present for this mark
	M1	Correctly shown substitution of limits, with a subtract sign between. The 3 need not be present for this mark, need not be simplified. *This mark can be implied if first M1 A1 awarded and final correct answer.* Allow a correct substitution of limits into any changed expression.
	A1	cao oe

Type MS · Turning sums of trig terms into products, and products into sums

Answer 5 of 8 · Question 6 part a · 2 marks

2020 NOV · Paper 01 (regional R) · extract 2 of 2 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Turning sums of trig terms into products, and products into sums

Answer 6 of 8 · Question 6 part b · 1 mark

2020 NOV · Paper 01 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

Question number		
6 (a)	$(\sin A\cos B+\cos A\sin B)+(\sin A\cos B-\cos A\sin B)$ $=2\sin A\cos B$ *	M1 A1 cso (2)
(b)	$\sin 8x+\sin 6x$	B1 (1)
(c)	$3\int_0^{\frac{\pi}{4}}(\sin 8x+\sin 6x)\,dx$ $=(3)\left[-\frac{1}{8}\cos 8x-\frac{1}{6}\cos 6x\right]_0^{\frac{\pi}{4}}$ $=(3)\left[\left(-\frac{1}{8}-0\right)-\left(-\frac{1}{8}-\frac{1}{6}\right)\right]$ $=\frac{1}{2}$ cao oe	M1 A1 M1 A1cao (4) [7]

Part	Mark	Additional Guidance
(a)	M1	Correct expression show.
	A1	cso
(b)	B1	For the expression shown
(c)	M1	$k\int_0^{\frac{\pi}{4}}(\sin 8x+\sin 6x)\,dx$ $k\neq 0$ or 1 k must be an integer. This mark can be awarded if the limits aren't seen on the integral.
	A1	Correctly integrated, the 3 and the limits need not be present for this mark
	M1	Correctly shown substitution of limits, with a subtract sign between. The 3 need not be present for this mark, need not be simplified. *This mark can be implied if first M1 A1 awarded and final correct answer.* Allow a correct substitution of limits into any changed expression.
	A1	cao oe

Type MS · Turning sums of trig terms into products, and products into sums

Answer 6 of 8 · Question 6 part b · 1 mark

2020 NOV · Paper 01 (regional R) · extract 2 of 2 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Turning sums of trig terms into products, and products into sums

Answer 7 of 8 · Question 10 part a · 2 marks

2023 JAN · Paper 01 (regional R) · extract 1 of 4 · find part a in this mark scheme extract

Question	Scheme	Marks
10(a)	$\cos(A-B)=\cos A\cos B+\sin A\sin B$ $\cos(A+B)=\cos A\cos B-\sin A\sin B$ $\cos(A-B)-\cos(A+B)=\sin A\sin B-(-\sin A\sin B)=2\sin A\sin B$ *	M1A1cso [2]
(b)	$[A-B=5\theta,\ A+B=9\theta\Rightarrow A=7\theta,\ B=2\theta]$ $\cos 5\theta-\cos 9\theta=2\sin 7\theta\sin 2\theta$ *	B1 cso [1]
(c)	$\sqrt{3}\sin 7\theta=2\sin 7\theta\sin 2\theta\Rightarrow 0=2\sin 7\theta\sin 2\theta-\sqrt{3}\sin 7\theta$ $0=\sin 7\theta\left(2\sin 2\theta-\sqrt{3}\right)\Rightarrow \sin 7\theta=0,\ 2\sin 2\theta-\sqrt{3}=0$ $\sin 7\theta=0\Rightarrow 7\theta=0,\ \pi,\ 2\pi\Rightarrow\theta=\frac{\pi}{7},\ \frac{2\pi}{7}$ $2\sin 2\theta-\sqrt{3}=0\Rightarrow\sin 2\theta=\frac{\sqrt{3}}{2}\Rightarrow 2\theta=\frac{\pi}{3},\ \frac{2\pi}{3}\Rightarrow\theta=\frac{\pi}{6},\ \frac{\pi}{3}$	M1A1 M1 M1A1 M1A1 [7]
(d)	$\tan 2x=\frac{\sin 2x}{\cos 2x}\Rightarrow\tan 2x\cos 2x=\sin 2x$ $\int_0^{\frac{\pi}{7}}8\sin 7x\sin 2x\,\mathrm{d}x=\left[\int_0^{\frac{\pi}{7}}4\times(2\sin 7x\sin 2x)\,\mathrm{d}x\right]$ $\int_0^{\frac{\pi}{7}}4(\cos 5x-\cos 9x)\,\mathrm{d}x=4\left[\frac{\sin 5x}{5}-\frac{\sin 9x}{9}\right]_0^{\frac{\pi}{7}}$ $4\left[\frac{\sin 5x}{5}-\frac{\sin 9x}{9}\right]_0^{\frac{\pi}{7}}=4\left[\left(\frac{\sin 5\times\frac{\pi}{7}}{5}-\frac{\sin 9\times\frac{\pi}{7}}{9}\right)-\left(\frac{\sin 0}{5}-\frac{\sin 0}{9}\right)\right]$# $=0.9729...\approx 0.973$	M1 M1 M1M1 M1A1 [6]
		Total 16 marks

Type MS · Turning sums of trig terms into products, and products into sums

Answer 7 of 8 · Question 10 part a · 2 marks

2023 JAN · Paper 01 (regional R) · extract 2 of 4 · find part a in this mark scheme extract

Question	Notes	Marks
10(a)	From the formula sheet $\cos(A-B)=\cos A\cos B+\sin A\sin B$ $\cos(A+B)=\cos A\cos B-\sin A\sin B$ Subtracts the two equations to give: $\cos(A-B)-\cos(A+B)=\sin A\sin B-(-\sin A\sin B)$	M1
	For the correct identity as shown with no errors, $\cos(A-B)-\cos(A+B)=2\sin A\sin B$ *	A1 cso [2]
(b)	For finding the value of A and the value of B $A-B=5\theta,\quad A+B=9\theta$ $\Rightarrow A=7\theta,\ B=2\theta$ Or as a minimum: $\cos(7\theta-2\theta)-\cos(7\theta+2\theta)=2\sin 7\theta\sin 2\theta$ *	B1cso [1]
(c)	Sets $\sqrt{3}\sin 7\theta=2\sin 7\theta\sin 2\theta$	M1
	Achieves the correct equation allow the terms in any order. $0=2\sin 7\theta\sin 2\theta-\sqrt{3}\sin 7\theta$	A1
	Factorises their equation $0=\sin 7\theta\left(2\sin 2\theta-\sqrt{3}\right)\Rightarrow \sin 7\theta=0,\ 2\sin 2\theta-\sqrt{3}=0$	M1
	For finding at least **one** correct value for θ using $\sin 7\theta=0$ $\sin 7\theta=0\Rightarrow 7\theta=0,\ \pi,\ 2\pi\Rightarrow\theta=\frac{\pi}{7},\ \frac{2\pi}{7}$	M1
	For both correct values $\theta=\frac{\pi}{7},\ \frac{2\pi}{7}$	A1
	For finding **one** correct value of θ using $2\sin 2\theta-\sqrt{3}=0$ $2\sin 2\theta-\sqrt{3}=0\Rightarrow\sin 2\theta=\frac{\sqrt{3}}{2}\Rightarrow 2\theta=\frac{\pi}{3},\ \frac{2\pi}{3}\Rightarrow\theta=\frac{\pi}{6},\ \frac{\pi}{3}$	M1
	For both correct values $\theta=\frac{\pi}{6},\ \frac{\pi}{3}$	A1 [7]
(d)	Uses the identity for $\tan 2x=\frac{\sin 2x}{\cos 2x}\Rightarrow\tan 2x\cos 2x=\sin 2x$	M1
	Substitutes the above into $8\sin 7x\cos 2x\tan 2x$ to give $\int_0^{\frac{\pi}{7}}8\sin 7x\sin 2x\,\mathrm{d}x=\left[\int_0^{\frac{\pi}{7}}4\times(2\sin 7x\sin 2x)\,\mathrm{d}x\right]$ Ignore integral sign and limits for this mark.	M1
	For substituting $\cos 5x-\cos 9x$ for $2\sin 7x\sin 2x$ to give $\int_0^{\frac{\pi}{7}}4(\cos 5x-\cos 9x)\,\mathrm{d}x$ Ignore integral sign and limits for this mark.	M1

Type MS · Turning sums of trig terms into products, and products into sums

Answer 7 of 8 · Question 10 part a · 2 marks

2023 JAN · Paper 01 (regional R) · extract 3 of 4 · find part a in this mark scheme extract

	Integrates $\int_0^{\frac{\pi}{7}} 4(\cos 5x - \cos 9x)\,dx = 4\left[\frac{\sin 5x}{5} - \frac{\sin 9x}{9}\right]_0^{\frac{\pi}{7}}$ Ignore limits for this mark. As a minimum they must obtain: $\left(\pm\frac{\sin 5x}{5} \pm \frac{\sin 9x}{9}\right)$ for the integration.	M1
	Substitutes the limits the correct way around $4\left[\frac{\sin 5x}{5} - \frac{\sin 9x}{9}\right]_0^{\frac{\pi}{7}} = 4\left[\left(\frac{\sin 5\times\frac{\pi}{7}}{5} - \frac{\sin 9\times\frac{\pi}{7}}{9}\right) - \left(\frac{\sin 0}{5} - \frac{\sin 0}{9}\right)\right]$ $= (0.9729...)$	M1
	For the correct value of $\int_0^{\frac{\pi}{7}} 8\sin 7x\cos 2x\tan 2x\,dx = 0.973$	A1 [6]
		Total 16 marks

Type MS · Turning sums of trig terms into products, and products into sums

Answer 7 of 8 · Question 10 part a · 2 marks

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Answer 8 of 8 · Question 10 part b · 1 mark

2023 JAN · Paper 01 (regional R) · extract 1 of 4 · find part b in this mark scheme extract

Question	Scheme	Marks
10(a)	$\cos(A-B)=\cos A\cos B+\sin A\sin B$ $\cos(A+B)=\cos A\cos B-\sin A\sin B$ $\cos(A-B)-\cos(A+B)=\sin A\sin B-(-\sin A\sin B)=2\sin A\sin B$ *	M1A1cso [2]
(b)	$[A-B=5\theta,\ A+B=9\theta\Rightarrow A=7\theta,\ B=2\theta]$ $\cos 5\theta-\cos 9\theta=2\sin 7\theta\sin 2\theta$ *	B1 cso [1]
(c)	$\sqrt{3}\sin 7\theta=2\sin 7\theta\sin 2\theta\Rightarrow 0=2\sin 7\theta\sin 2\theta-\sqrt{3}\sin 7\theta$ $0=\sin 7\theta\left(2\sin 2\theta-\sqrt{3}\right)\Rightarrow \sin 7\theta=0,\ 2\sin 2\theta-\sqrt{3}=0$ $\sin 7\theta=0\Rightarrow 7\theta=0,\ \pi,\ 2\pi\Rightarrow\theta=\frac{\pi}{7},\ \frac{2\pi}{7}$ $2\sin 2\theta-\sqrt{3}=0\Rightarrow\sin 2\theta=\frac{\sqrt{3}}{2}\Rightarrow 2\theta=\frac{\pi}{3},\ \frac{2\pi}{3}\Rightarrow\theta=\frac{\pi}{6},\ \frac{\pi}{3}$	M1A1 M1 M1A1 M1A1 [7]
(d)	$\tan 2x=\frac{\sin 2x}{\cos 2x}\Rightarrow\tan 2x\cos 2x=\sin 2x$ $\int_0^{\frac{\pi}{7}}8\sin 7x\sin 2x\,\mathrm{d}x=\left[\int_0^{\frac{\pi}{7}}4\times(2\sin 7x\sin 2x)\,\mathrm{d}x\right]$ $\int_0^{\frac{\pi}{7}}4(\cos 5x-\cos 9x)\,\mathrm{d}x=4\left[\frac{\sin 5x}{5}-\frac{\sin 9x}{9}\right]_0^{\frac{\pi}{7}}$ $4\left[\frac{\sin 5x}{5}-\frac{\sin 9x}{9}\right]_0^{\frac{\pi}{7}}=4\left[\left(\frac{\sin 5\times\frac{\pi}{7}}{5}-\frac{\sin 9\times\frac{\pi}{7}}{9}\right)-\left(\frac{\sin 0}{5}-\frac{\sin 0}{9}\right)\right]$# $=0.9729...\approx 0.973$	M1 M1 M1M1 M1A1 [6]
		Total 16 marks

Type MS · Turning sums of trig terms into products, and products into sums

Answer 8 of 8 · Question 10 part b · 1 mark

2023 JAN · Paper 01 (regional R) · extract 2 of 4 · find part b in this mark scheme extract

Question	Notes	Marks
10(a)	From the formula sheet $\cos(A-B)=\cos A\cos B+\sin A\sin B$ $\cos(A+B)=\cos A\cos B-\sin A\sin B$ Subtracts the two equations to give: $\cos(A-B)-\cos(A+B)=\sin A\sin B-(-\sin A\sin B)$	M1
	For the correct identity as shown with no errors, $\cos(A-B)-\cos(A+B)=2\sin A\sin B$ *	A1 cso [2]
(b)	For finding the value of A and the value of B $A-B=5\theta,\quad A+B=9\theta$ $\Rightarrow A=7\theta,\ B=2\theta$ Or as a minimum: $\cos(7\theta-2\theta)-\cos(7\theta+2\theta)=2\sin 7\theta\sin 2\theta$ *	B1cso [1]
(c)	Sets $\sqrt{3}\sin 7\theta=2\sin 7\theta\sin 2\theta$	M1
	Achieves the correct equation allow the terms in any order. $0=2\sin 7\theta\sin 2\theta-\sqrt{3}\sin 7\theta$	A1
	Factorises their equation $0=\sin 7\theta\left(2\sin 2\theta-\sqrt{3}\right)\Rightarrow \sin 7\theta=0,\ 2\sin 2\theta-\sqrt{3}=0$	M1
	For finding at least **one** correct value for θ using $\sin 7\theta=0$ $\sin 7\theta=0\Rightarrow 7\theta=0,\ \pi,\ 2\pi\Rightarrow\theta=\frac{\pi}{7},\ \frac{2\pi}{7}$	M1
	For both correct values $\theta=\frac{\pi}{7},\ \frac{2\pi}{7}$	A1
	For finding **one** correct value of θ using $2\sin 2\theta-\sqrt{3}=0$ $2\sin 2\theta-\sqrt{3}=0\Rightarrow\sin 2\theta=\frac{\sqrt{3}}{2}\Rightarrow 2\theta=\frac{\pi}{3},\ \frac{2\pi}{3}\Rightarrow\theta=\frac{\pi}{6},\ \frac{\pi}{3}$	M1
	For both correct values $\theta=\frac{\pi}{6},\ \frac{\pi}{3}$	A1 [7]
(d)	Uses the identity for $\tan 2x=\frac{\sin 2x}{\cos 2x}\Rightarrow\tan 2x\cos 2x=\sin 2x$	M1
	Substitutes the above into $8\sin 7x\cos 2x\tan 2x$ to give $\int_0^{\frac{\pi}{7}}8\sin 7x\sin 2x\,\mathrm{d}x=\left[\int_0^{\frac{\pi}{7}}4\times(2\sin 7x\sin 2x)\,\mathrm{d}x\right]$ Ignore integral sign and limits for this mark.	M1
	For substituting $\cos 5x-\cos 9x$ for $2\sin 7x\sin 2x$ to give $\int_0^{\frac{\pi}{7}}4(\cos 5x-\cos 9x)\,\mathrm{d}x$ Ignore integral sign and limits for this mark.	M1

Type MS · Turning sums of trig terms into products, and products into sums

Answer 8 of 8 · Question 10 part b · 1 mark

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	Integrates $\int_0^{\frac{\pi}{7}} 4(\cos 5x - \cos 9x)\,dx = 4\left[\frac{\sin 5x}{5} - \frac{\sin 9x}{9}\right]_0^{\frac{\pi}{7}}$ Ignore limits for this mark. As a minimum they must obtain: $\left(\pm\frac{\sin 5x}{5} \pm \frac{\sin 9x}{9}\right)$ for the integration.	M1
	Substitutes the limits the correct way around $4\left[\frac{\sin 5x}{5} - \frac{\sin 9x}{9}\right]_0^{\frac{\pi}{7}} = 4\left[\left(\frac{\sin 5\times\frac{\pi}{7}}{5} - \frac{\sin 9\times\frac{\pi}{7}}{9}\right) - \left(\frac{\sin 0}{5} - \frac{\sin 0}{9}\right)\right]$ $= (0.9729...)$	M1
	For the correct value of $\int_0^{\frac{\pi}{7}} 8\sin 7x\cos 2x\tan 2x\,dx = 0.973$	A1 [6]
		Total 16 marks

Type MS · Turning sums of trig terms into products, and products into sums

Answer 8 of 8 · Question 10 part b · 1 mark

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